FINAL YEAR PROJECT PROPOSAL

MoodBand: An Inclusive Wearable System for Real-Time Mood Detection and Adaptive Sensory Feedback

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Abstract

Emotional well-being is closely tied to mental health, productivity, and quality of life, yet many people — particularly those who are visually impaired — lack accessible, unobtrusive tools to monitor and regulate their own mood in real time. This proposal presents MoodBand, a system that integrates with a Fitbit wearable via the Google Health API to continuously sense physiological and motion signals — including heart rate, heart-rate variability, and activity — and infers the wearer's emotional state using a lightweight machine-learning classifier, run on-device where feasible or via a connected backend. Once a mood state (stress, sadness, or anger) is detected, the companion mobile application autonomously triggers a personalized feedback response delivered entirely through the phone: auditory (spoken prompts or calming audio through the phone speaker), haptic (phone-vibration patterns paced to guide breathing), or visual (in-app colour cues and a screen-reader-accessible dashboard) — chosen according to whether the user is sighted or visually impaired. The goal is to close the feedback loop between mood sensing and mood regulation in a way that is inclusive by design and that uses hardware people already own, rather than a custom-built device. This document outlines the background, problem statement, objectives, methodology, system architecture, implementation plan, timeline, and budget for the proposed final year project.

1. Introduction and Background

Wearable technology has moved from simple activity tracking toward richer physiological sensing, enabling continuous, passive insight into a user's physical and emotional state. Emotions influence decision-making, social interaction, and long-term mental health, yet most people have no simple way to notice a negative mood shift before it escalates into stress or withdrawal. Existing mood-tracking apps largely rely on manual self-reporting, which is easy to forget and inherently reactive rather than real-time.

Visually impaired individuals face an additional barrier: most consumer mood-tracking and well-being tools are built around visual interfaces — screens, color-coded charts, and icon-based dashboards — that are inaccessible without a screen reader. This creates a gap in inclusive well-being technology. A system that senses mood using signals from an existing wearable device such as a Fitbit, rather than through visual self-report or custom hardware, and responds through appropriately chosen sensory channels delivered via a mobile app can serve both sighted and visually impaired users from the same underlying platform, rather than requiring a separate accessible version as an afterthought.

2. Problem Statement

Current mood-monitoring solutions share three limitations that this project seeks to address:

- They are largely self-report based, requiring conscious, repeated user input rather than continuous passive sensing.
- They provide little or no closed-loop intervention — most only log or visualize mood after the fact, without acting to help the user regulate it in the moment.
- They are not designed with accessibility in mind, offering feedback almost exclusively through visual dashboards that exclude visually impaired users.

There is therefore a need for a wearable-integrated system that can detect negative mood states in real time and respond immediately with feedback appropriate to the user's sensory needs — visual for sighted users, and auditory or haptic for visually impaired users — with the explicit goal of nudging the user toward an improved emotional state.

3. Aim and Objectives

3.1 Aim

To design, develop, and evaluate a mood-detection system that leverages a Fitbit wearable to sense a user's physiological signals, classifies their emotional state in real time, and delivers adaptive auditory, haptic, or visual feedback through a companion mobile app intended to improve that mood, for use by both sighted and visually impaired individuals, on both Android and iOS.

3.2 Objectives

1. Review existing literature and commercial systems on affective computing, wearable physiological sensing, and accessible feedback technology.
2. Integrate with a Fitbit wearable via the Google Health API to pull heart-rate, heart-rate-variability, and related physiological/motion data in real time, on both Android and iOS.
3. Collect physiological data and train a machine-learning model capable of classifying mood into categories such as happy, neutral, sad, stressed, and angry.
4. Design a feedback decision module that selects an appropriate intervention (auditory, haptic, or visual) based on detected mood and user accessibility profile.
5. Implement the auditory, haptic, and visual feedback mechanisms within the companion mobile application, built to be accessible via screen readers.
6. Test and evaluate the system's mood-classification accuracy and the perceived effectiveness of its feedback on a sample of sighted and visually impaired participants.

4. Scope of the Project

MoodBand implements a four-stage sense → process → classify → feedback loop: physiological and motion data are collected from an existing Fitbit wearable via the Google Health API; that data is processed and classified into a discrete mood category by a trained machine-learning model; the classified mood is used to select an appropriate feedback intervention; and that feedback is delivered back to the user through their phone, closing the loop. Fitbit was selected over alternatives such as the Apple Watch specifically because its cloud-based API works identically regardless of the phone's operating system, allowing the companion app to support both Android and iOS users rather than being limited to one platform.

This scope defines the sense-to-feedback loop as a logical pipeline, not a mandated physical deployment. The processing (mood classification) and feedback-decision stages may run on the mobile device itself, on a connected laptop, or on a cloud backend (e.g., a lightweight cloud function), whichever location proves most practical given on-device performance constraints encountered during development. A fully on-device mobile implementation, where every stage runs locally on the phone, is the ideal target and the configuration the project will aim for by default — but it is not a strict requirement of this scope. What matters is that the complete sense-to-feedback loop functions end-to-end and in near real time, regardless of exactly which machine executes the processing and feedback-decision steps. The feedback-actuation stage itself (speaker, vibration motor, screen) necessarily runs on the user's phone, since that is the hardware the user directly experiences.

The scope includes detection of a limited set of discrete mood categories rather than a continuous emotional spectrum, and evaluation with a small user sample appropriate for a final year project timeline completed by a single student.

Design or fabrication of custom wearable hardware is outside the scope of this project, since an existing consumer device (a Fitbit wearable) is used for sensing instead. Also outside of scope: long-term clinical validation, mass-manufacturing considerations, and integration with third-party health platforms beyond the Google Health API.

A related but distinct idea — teaching or training visually impaired users to interpret colour-based mood cues directly (for example, learning a personal colour-to-mood association through a dedicated non-visual training modality) — is deliberately kept out of scope here. Within this project, visually impaired users are served by the auditory and haptic feedback channels described in Section 8, not by an adapted version of the colour cue. The

colour-decoding concept is recognised as a promising and non-trivial research direction in its own right and is proposed as a separate, follow-on project for a future project cycle rather than a deliverable of this one.

5. Literature Review

5.1 Affective Computing Fundamentals

Picard's foundational work [1] is where affective computing as a discipline really starts. Her central claim was that computers could be built to recognize and respond to emotional states, not just ignore them as noise. That idea sounds almost obvious now, but at the time it wasn't, and it's still the premise MoodBand rests on: that internal affective states leave measurable traces a system can learn to read.

Decades later, Conceição et al. [2] show what that premise looks like in practice with modern tools. They build on Russell's circumplex model, which maps emotion along two axes, valence and arousal, and use a deep-seeded clustering algorithm to classify emotional states from signals collected by a wearable wristband, testing it across three separate benchmark datasets. What makes this paper useful for MoodBand specifically is that it's working with the same kind of hardware form factor (a wrist-worn device) and the same underlying signal types we'd expect from a Fitbit, and it still manages to map those signals onto discrete emotional categories reasonably well.

Implication for MoodBand: together, [1] and [2] justify the project's core premise — that physiological signals can be mapped onto discrete emotional categories — and specifically justify grounding MoodBand's mood labels (happy, sad, stressed, angry, neutral) in the valence/arousal circumplex model rather than inventing an ad hoc labelling scheme. This directly shaped the mood categories used throughout the system design (Sections 6–8).

5.2 Wearable Physiological Sensing

Moving from theory to the specific question of wearables, Saganowski et al.'s systematic review [3] is a useful starting point. They find that heart rate, HRV, and electrodermal activity dominate the literature as the go-to signals for everyday emotion recognition, and they flag how much harder things get once you move out of the lab and into real, everyday conditions. That's exactly the setting MoodBand is targeting.

On the consumer-hardware side specifically, [4] is probably the single most reassuring paper on this list. Using the LifeSnaps dataset, built from actual Fitbit Sense smartwatches rather than research-grade sensors, the authors predict anxiety from everyday wearable data and get results suggesting a plain consumer device carries useful signal, not just noise.

That said, [5] is a good reality check on the same point. The authors ran into some genuinely annoying constraints when pulling Fitbit data for ML purposes: step counts reading zero while the device is just charging, HR data missing when the device isn't worn properly, all requiring manual filtering. Most notably, Fitbit only computes HRV while the user is asleep and relatively still, which would otherwise mean HRV could not be relied upon as a live, daytime signal.

Implication for MoodBand: papers [4] and [5] both use Fitbit devices specifically, which is directly relevant since Fitbit is also the device selected for this project (see Section 7.2 / Section 5.7). The choice of Fitbit was driven primarily by a requirement outside the scope of this literature review on its own: the companion app must work for both Android and iOS users. Fitbit's cloud-based API serves data identically regardless of the user's phone platform, whereas device-native alternatives such as Apple's HealthKit only work on iOS. This means the HRV constraint identified in [5] — that Fitbit only computes HRV overnight — carries over directly into MoodBand's design, and the classification pipeline (Section 7.3) is built around that constraint rather than around it being designed away.

5.3 Mood/Emotion Classification Models

Once the signals are available, the question is what model actually does the classifying. The systematic review in [6] gives a pretty clear answer: classical machine learning methods, Random Forest and Support Vector Machines in particular, show up in the large majority of wearable-based stress detection studies, with Random Forest especially holding up well on EDA and HRV features.

[7] adds two things on top of that. First, the authors design their classification approach specifically for resource-constrained, on-device deployment — directly relevant given MoodBand's classifier must run on a mobile phone rather than a server. Second, they compare personalized models against generalized ones and find personalized models can win out even with limited data per person, which matters given MoodBand's own user trial will necessarily be small.

Implication for MoodBand: [6] directly influenced the choice of classifier family — Random Forest is used as the primary candidate classifier (Section 7.3) specifically because it is the best-performing classical model on EDA/HRV-style wearable features in the reviewed literature, rather than defaulting to a heavier deep-learning approach that would be harder to justify running on a phone. [7] influenced two further decisions: favouring lightweight, resource-constrained model architectures suitable for on-device or near-device inference (Section 9), and adopting a personalized-model evaluation strategy alongside a generalized one, given the project's necessarily small participant sample (Section 7.6).

5.4 Accessible Feedback Design

The last piece of MoodBand, getting feedback to the user through non-visual channels, is backed by a smaller but still useful set of papers. The iFeeling system [8] is about as close a precedent as exists: it takes a detected emotional state and renders it as a distinct vibration pattern on a phone, built explicitly with visually impaired users in mind.

[9] widens the lens a bit. It's a recent, large-scale systematic literature review spanning two decades (2004 to 2024) of haptic assistive tools for blind and low-vision users, drawing on 132 papers, and it's useful less for any single finding and more as a way to check iFeeling's design choices against the broader literature.

Finally, [10] grounds all of this in actual user experience. Yeo and Chew interviewed visually impaired people in Singapore about how they use mobile apps day to day, and the picture isn't entirely rosy: screen readers often don't work as reliably as one would hope. Their recommendations explicitly call for combining audio and haptic feedback rather than relying on either alone, and for taking users' emotional wellbeing seriously as a design goal.

Implication for MoodBand: [8] directly shaped the haptic feedback channel — MoodBand's vibration-based mood cues (Section 7.4, Section 8) are modelled on iFeeling's approach of rendering a detected emotional state as a distinct phone-vibration pattern. [9] broadened this into a fuller design vocabulary: MoodBand's use of graduated or directional vibration patterns (e.g., descending vibration intensity for anger, see Table 1) draws on patterns documented across this wider body of research rather than the single iFeeling precedent alone. [10] most directly shaped a specific design rule: because screen readers were found to behave unreliably in practice, MoodBand pairs every visual cue with a corresponding audio or haptic equivalent by default (Section 7.4), rather than treating a screen-reader-compatible UI as sufficient on its own.

5.5 Synthesis

Put together, the pieces MoodBand depends on each have reasonable support behind them. Physiological signals can be mapped onto discrete moods [1], [2]. A consumer wearable like Fitbit is a workable, if imperfect, way to collect those signals [3]–[5]. Classical, lightweight models are a sensible fit for the classification step given the deployment constraints [6], [7]. And haptic feedback in particular has real precedent for communicating affective state to visually impaired users [8]–[10]. What doesn't exist yet, as far as this review found, is a single system that ties all four pieces together into one mobile-first pipeline — a gap this project aims to address.

5.6 Sensor Recommendations Across Reviewed Literature

Table 2 maps each reviewed paper to the physiological sensors or signals it used or recommended, alongside the sensors MoodBand will actually move forward with.

Paper	Sensors Used / Recommended
[1] Picard (1997)	N/A — conceptual/foundational text, no specific sensor
[2] Conceição et al.	Heart rate / blood-volume pulse (PPG), electrodermal activity (EDA), skin temperature (via Empatica E4 wristband)

Paper	Sensors Used / Recommended
[3] Saganowski et al. (review)	Heart rate, HRV, electrodermal activity (identified as the dominant signals across the wearable-sensing literature)
[4] Velu et al. (LifeSnaps/Fitbit)	Heart rate, Fitbit-derived stress score, sleep, step count (Fitbit Sense)
[5] King et al.	Heart rate, HRV (sleep-only), step count / motion (Fitbit intraday data)
[6] Pataca et al. (review)	Electrodermal activity, HRV (found Random Forest performs best specifically on these two)
[7] Irfan et al.	ECG (minimal wearable sensing)
[8] iFeeling	N/A — haptic output/rendering paper, not a sensing paper
[9] Jiang et al. (review)	N/A — haptic feedback review, not sensing
[10] Yeo & Chew	N/A — qualitative UX study, not sensing
Sensors MoodBand will use	**Heart rate (continuous, real-time) and motion/accelerometer (continuous), with heart-rate variability (HRV) incorporated as a supplementary overnight-only signal**

Table 2: Sensors recommended per paper vs. sensors selected for MoodBand

These are the signals consistently available through a consumer-grade, cross-platform-friendly wearable API (Fitbit, via the Google Health API). Electrodermal activity and skin temperature — used in the lab-grade Empatica E4 wristband in [2] and recommended generally in [3] and [6] — are not exposed by any mainstream consumer wearable's public API and are therefore out of reach for this project. ECG-based sensing as explored in [7] likewise requires dedicated ECG-capable hardware (e.g., a chest strap, or Apple Watch's ECG app) not available through Fitbit's consumer API. MoodBand's sensor selection therefore reflects the intersection of what the literature recommends and what a consumer-grade, cross-platform wearable API actually exposes, with HRV demoted to a supplementary overnight signal per the constraint identified in [5].

5.7 Device Selection Summary

To select a specific wearable, four candidate devices — Fitbit, Apple Watch, Samsung Galaxy Watch, and WHOOP — were compared against price, cross-platform support, developer API accessibility, real-time data availability, and development-environment fit. The full comparison is documented separately (MoodBand — Wearable Device Selection: Comparative Analysis, submitted for supervisor review); this section summarizes that analysis and its conclusion.

One piece of context shaped the timing of this decision: the legacy Fitbit Web API (the one referenced by [4] and [5]) was retired by Google — developer support ended September 30, 2026, with full shutdown October 30, 2026 — and replaced by the new Google Health API. This is a net positive for the project: the legacy API required a justification request to unlock sub-daily heart-rate data, whereas the new API provides 5-second-resolution heart rate by default with no special approval step.

Criterion	Fitbit	Apple Watch (SE 3)	Samsung Galaxy Watch	WHOOP
Price (new / used)	$70–130 / $30–80	$220–250 / $120–180	$170–310 / $100–150	Bundled with membership
Works on Android?	Yes	No — iPhone-only, structurally	Yes	Yes

Criterion	Fitbit	Apple Watch (SE 3)	Samsung Galaxy Watch	WHOOP
Works on iOS?	Yes	Yes	No — continuous streaming is Android-only	Yes
API approval needed?	No (new Google Health API)	No (HealthKit)	Only for Samsung's cloud API (1–2 wks)	Free API, but returns no data without paid membership
Real-time daytime HR?	Yes — 5-sec resolution	Yes (workout mode)	Yes — on-device Sensor SDK	No — no continuous/raw HR via API
Matches lit. review?	Yes — [4], [5] used Fitbit directly	No direct precedent	No direct precedent	No direct precedent

Table 3: Condensed device comparison (full analysis in the separate device-selection document)

Decision: Fitbit, integrated via the Google Health API, was finalized as the project's sensing device. The deciding factor was the hard requirement that the companion app support both Android and iOS: this structurally rules out Apple Watch (pairs with iPhone only) and Samsung Galaxy Watch (continuous real-time streaming is Android-only via the Samsung Health Sensor SDK), regardless of their other technical merits. Fitbit's cloud-based API is the only option on the shortlist that serves identical data regardless of phone platform. WHOOP was ruled out independently of the platform question, since its API exposes no continuous or raw heart-rate data under any circumstance and requires an active paid membership just to return data at all.

6. Proposed System Overview

The proposed system, MoodBand, is organized into four functional layers, implementing the sense → process → classify → feedback loop described in Section 4: a sensing layer that pulls raw physiological and motion signals from the user's Fitbit device via the Google Health API; a processing layer that cleans the signal and classifies mood using a trained model; a decision layer that maps the detected mood and the user's accessibility profile to an appropriate intervention; and a feedback layer that actuates the chosen auditory, haptic (via phone vibration), or visual response. The processing and decision layers are implemented in software that can run on the mobile device, a connected laptop, or a cloud backend; full on-device mobile execution is the target end-state, but the architecture does not require every stage to run on the phone. The feedback/actuation layer, by contrast, necessarily runs on the phone, since it drives hardware (speaker, vibration motor, screen) the user directly experiences. Because the sensing layer communicates with Fitbit's cloud API rather than a platform-specific on-device framework, this architecture functions identically whether the companion app runs on Android or iOS. Figure 1 illustrates this architecture.

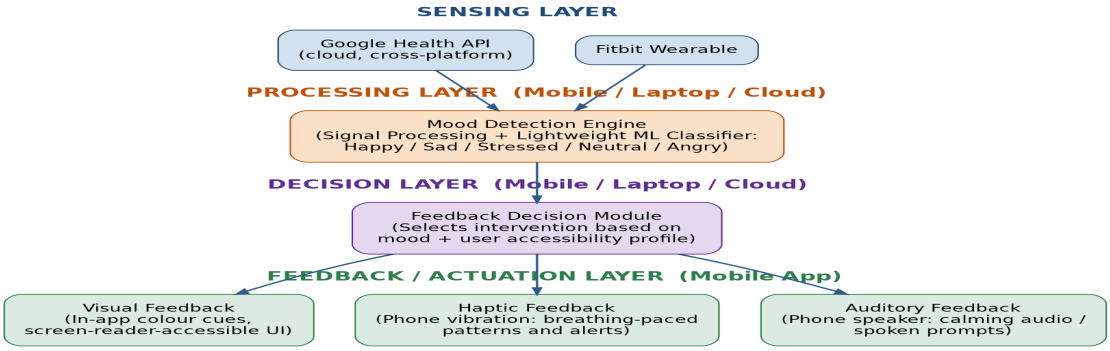


Figure 1: Proposed system architecture of MoodBand

7. Methodology

Given that this is a single-student project with a fixed 16-week timeline, the system will be built and validated in two phases rather than attempting live wearable integration from day one. In the first phase, the mood-classification model, feedback-decision logic, and mobile companion app are built and tested end-to-end using publicly available Fitbit and Apple Watch datasets (see Section 5.6) as a stand-in for live sensor data, in order to validate the full sense-to-feedback pipeline and surface design issues early. Only once this offline version is working and tested does the second phase begin: integrating the live Fitbit sensing layer via the Google Health API (Section 7.2) in place of the offline dataset. This ordering is reflected in the project timeline (Section 11).

7.1 Fitbit Integration

Rather than building custom wearable hardware, MoodBand integrates with an existing Fitbit wearable via the Google Health API (the successor to the legacy Fitbit Web API, which Google is retiring in 2026). Fitbit was selected over device-native alternatives such as the Apple Watch specifically because its API is cloud-based rather than tied to a single phone operating system: the same REST calls return data whether the companion app is running on Android or iOS, which is a hard requirement for this project. This provides access to heart rate, heart-rate variability, and motion data without requiring custom sensor design, fabrication, or a dedicated charging and power-management system — a necessary adaptation given the project is being completed by a single student within a 16-week timeline. The literature review (Section 5.2) supports this approach: consumer wearables, and Fitbit specifically, have been shown to carry meaningful signal for affective-state inference, even with some constraints (e.g., HRV only being available overnight) that must be designed around.

7.2 Data Collection and Mood Classification

Model development begins with publicly available Fitbit and Apple Watch datasets (Section 5.6), used to train and validate an initial classifier before any live integration work begins. Once the offline model and pipeline are validated, labelled data from the project's own small-scale user trial will be gathered through a structured protocol in which volunteer participants are exposed to mood-inducing stimuli (e.g., calming music, stressful timed tasks, or neutral video) while Fitbit-derived signals are logged via the Google Health API, and self-reported mood labels are collected as ground truth. Standard signal-processing techniques (filtering, normalization, and feature extraction from heart-rate and motion data) will be applied before training a lightweight classifier — such as a Random Forest, Support Vector Machine, or small neural network — selected based on accuracy versus resource constraints, consistent with findings from the literature review on wearable-based stress and emotion classification (Section 5.3). Because HRV is only reliably available from Fitbit overnight, the real-time, waking-state classification pipeline relies primarily on heart rate and motion features, with HRV incorporated only as a supplementary nightly signal.

7.3 Feedback Decision Logic

A rule-based decision module will map each classified mood, together with a stored user accessibility profile (sighted or visually impaired), to a specific feedback routine. For example, a detected stress state for a visually impaired user triggers a slow phone-vibration pattern paced to guide breathing, accompanied by an optional spoken prompt through the phone speaker, whereas the same state for a sighted user triggers a calming colour animation on the app screen alongside the option to view breathing guidance visually.

7.4 Companion Mobile Application

As the sole student on this project, the originally proposed custom wearable prototype has been descoped in favour of leveraging an existing Fitbit wearable for sensing, with the companion mobile application responsible for the user-facing feedback cycle. The app will allow users to view mood history, adjust feedback preferences, and configure their accessibility profile, in addition to delivering all auditory, haptic, and visual feedback directly through the phone's built-in speaker, vibration motor, and screen. The app will be built with accessibility guidelines in mind (screen-reader compatibility, high-contrast mode, voice control) to ensure it is usable by both sighted and visually impaired users.

7.5 Testing and Evaluation

The system will be evaluated in two stages: first, technical evaluation of mood-classification accuracy against the labelled dataset using standard metrics (accuracy, precision, recall, F1-score); second, a small-scale user study with both sighted and visually impaired volunteers to assess the usability of the feedback mechanisms and gather qualitative feedback on perceived mood improvement, informed by consent and ethical approval procedures appropriate to human-subjects research.

8. Feedback Channels by User Type

Table 4 summarizes representative feedback responses the system will provide for each detected mood, differentiated by user accessibility profile. All feedback is delivered through the user's phone (speaker, vibration motor, and screen) rather than dedicated wearable hardware.

Mood Detected	Feedback for Sighted Users	Feedback for Visually Impaired Users
Stress / Anxiety	Cool-toned in-app colour pulse + on-screen breathing animation	Slow-paced phone vibration pattern synced with spoken breathing cues
Sadness	Warm colour cue + suggested uplifting playlist on app	Gentle spoken affirmation + calming audio track
Anger / Frustration	Red-to-green fading in-app colour cue as a calming timer	Descending vibration intensity + guided calming voice prompt
Neutral / Happy	Soft green confirmation banner, no interruption	Brief pleasant chime, no interruption

Table 4: Example mood-to-feedback mapping by user type

9. Tools and Technologies

- Wearable Integration: Google Health API (Fitbit — cloud-based, works identically on Android and iOS; no custom hardware required)
- Machine Learning: Python, scikit-learn / TensorFlow Lite for lightweight inference
- Processing/Backend (optional): if on-device performance constraints require it, the mood-classification and feedback-decision logic can run on a lightweight cloud backend (e.g., a Firebase Cloud Function) or a connected laptop instead of fully on-device; full on-device execution remains the target end state (see Section 4)
- Mobile Application: Flutter or React Native (chosen for built-in accessibility widgets and cross-platform Android/iOS support from a single codebase); delivers all auditory, haptic, and visual feedback directly from the phone
- Data Storage: Local on-device buffer with cloud sync (Firebase) for mood history
- Version Control & Documentation: Git/GitHub, Figma (app UI/UX mockups)

10. Expected Outcomes

- Wearable-Integrated Mobile App: A functional mobile application, working on both Android and iOS, that connects to a Fitbit wearable via the Google Health API, pulls physiological and motion data in real time, and runs the mood-classification pipeline on-device or via a connected backend.
- Mood-Classification Model: A trained machine-learning model capable of classifying mood into at least four distinct categories (e.g., happy, neutral, sad, stressed, angry), first validated on public datasets and then on data sourced from the wearable.

- In-App Feedback Subsystem: A working feedback module delivering distinguishable auditory, haptic (phone vibration), and visual interventions, selected according to detected mood and the user's accessibility profile.
- Accessible Mobile Interface: A companion app interface accessible to both sighted and visually impaired users, including full screen-reader compatibility.
- Usability Evaluation Report: A report summarizing mood-classification performance and participant feedback from user trials involving sighted and visually impaired participants.
- Documented Architecture: A documented, extensible architecture that can serve as a foundation for future clinical-grade or commercial development.

11. Project Timeline

The project is planned over a 16-week period, summarised in Table 5. As described in Section 7, the model, feedback module, and mobile app are built and tested first using offline, publicly available datasets; Fitbit / Google Health API integration is scheduled after mobile app development, once the full product has been validated to work end-to-end.

Activity	Wk 1–2	Wk 3–6	Wk 7–10	Wk 11–13	Wk 14–16
Literature review & requirements gathering	X				
Dataset acquisition (Fitbit/Apple Watch datasets) & preprocessing	X	X			
Mood-classification model training (offline, public datasets)		X	X		
Feedback module (audio/haptic/visual) development		X	X		
Mobile companion app development			X	X	
Offline system integration & testing (app + model + feedback, no live wearable)				X	
Fitbit / Google Health API integration & setup				X	X
End-to-end testing with live wearable data					X
User trials (sighted & visually impaired)					X
Evaluation, documentation & final report					X

Table 5: Proposed project timeline

12. Budget Estimate

As the project now integrates with an existing Fitbit wearable rather than building custom hardware, the required budget is minimal compared to the original prototype-based plan. Table 6 summarises the estimated costs.

Item	Estimated Cost
Fitbit wearable (Charge 6 or Inspire 3) — new or used	$30–$130 (if not already owned)
Google Health API access (successor to the Fitbit Web API)	$0 (free for development use)
Mobile app development tools (Flutter/React Native, Figma)	$0 (free tiers)
Cloud storage/sync (Firebase)	$0 (free tier, sufficient for project scale)
Miscellaneous (testing devices for both Android and iOS, participant incentives)	$50
Contingency (10%)	$5
Total Estimated Cost (excluding the Fitbit device, if not already available)	**~$55**

Table 6: Estimated project budget

13. Significance of the Project

This project contributes to two growing fields simultaneously: wearable-integrated affective computing and inclusive assistive technology. Rather than treating accessibility as an add-on, MoodBand is designed from the outset to serve sighted and visually impaired users through a single adaptable platform, offering a model for how future consumer well-being apps might be built for inclusivity by default. By building on existing wearable hardware rather than a custom device, the project also demonstrates a practical, low-barrier path to deploying accessible affective computing using devices people already own. If successful, the prototype could serve as a foundation for applications in stress management, mental-health support tools, and assistive technology for independent living.

14. Conclusion

This proposal has outlined the motivation, objectives, methodology, and implementation plan for MoodBand, a system that integrates with an existing Fitbit wearable to detect mood in real time, on both Android and iOS, and respond with accessible, adaptive feedback aimed at improving the user's emotional state. By combining wearable-sourced physiological sensing, lightweight machine learning, and accessibility-conscious feedback design, built and validated offline before live wearable integration, the project addresses a genuine gap in current well-being technology while remaining feasible for a single student to complete within a final year project timeline and budget.

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